PubMed/PMC Task 2: Open-Ended Exploratory Questions

Task 2 is designed around broad, exploratory biomedical research inquiries, often requiring multi-document thematic synthesis. Systems under challenge must generate a clear, natural language response that is systematically grounded in valid PubMed and PubMed Central (PMC) citations, backed by explicit supporting passages.

To ensure a standardized evaluation, systems must use evidence *exclusively* from the official challenge corpus comprised of a subset of PubMed abstracts and PMC Open-Access full-text articles. Consequently, all reference responses must cite documents contained strictly within this challenge corpus. The ideal response is highly objective and concise - synthesizing all essential facts without extraneous background filler - and must explicitly highlight areas of scientific uncertainty, mixed evidence, or competing hypotheses in the literature, if applicable.

The following is an example of a question and the corresponding expected response for this Challenge Task:

Question: What genetic and other molecular changes are associated with the acquisition of treatment resistance leading to disease progression in chronic lymphocytic leukemia (CLL) patients treated with ibrutinib?

Expected Response:

Ibrutinib (aka PCI-32765) inhibits a number of protein tyrosine kinases, including Bruton's Tyrosine Kinase (BTK), through covalent bonding to cysteine residues in their active sites, and has been approved for the treatment of chronic lymphocytic leukemia (CLL), where signaling through the B cell antigen receptor mediated by BTK is required for cell activation and survival [1]. Using targeted and whole-exome sequencing and clonal evolution analysis, mutations in the BTK, PLCG2, and ITPKB genes were associated with clonal expansion in different patients with relapsing disease, suggesting the acquisition of treatment resistance due to mutations in these genes [2].

References

[1] Ahn IE, Brown JR. Targeting Bruton's Tyrosine Kinase in CLL. Frontiers in Immunology. 2021. PMID: 34248972, PMCID: PMC8261291.

[2] Landau DA, Sun C, Rosebrock D, et al. The evolutionary landscape of chronic lymphocytic leukemia treated with ibrutinib targeted therapy. Nature Communications. 2017. PMID: 29259203, PMCID: PMC5736707.

Challenge Dataset

This challenge comes with its own Challenge Corpus that can be used in responding to the questions in this task. It consists of two constituent corpora:

PubMed

- The PubMed baseline files for the challenge are available at https://ftp.ncbi.nlm.nih.gov/pubmed/.pubmed-spark/
- This corpus includes all citations with a publication date from January 1,1959 to July 31,2026 and excludes any citation with the publication type "preprint" (MeSH UI D000076942)
- For more information about this dataset, see the relevant PubMed documentation page (https://pubmed.ncbi.nlm.nih.gov/download/).

PubMed Central (PMC)

- A subset of PMC shall be used as a source of full-text documents for responding to questions in this task. The resulting subset includes PMC publications that satisfy the specified Creative Commons licensing, NIH funding or intramural status, publication-date, and publication-status criteria.
- A list of PMCIDs defining the PMC component of the Challenge Corpus is available in the file named "**Challenge_PMCIDs.txt**"
- The corresponding PMC articles may be downloaded or accessed programmatically using NCBI/PMC-supported services, including the PMC Cloud Service, PMC OAI-PMH API, BioC API, NCBI E-Utilities, and PMC ID Converter API. The PMC Cloud Service supports bulk access to machine-readable article files, while the OAI-PMH and BioC APIs support programmatic retrieval of article content. See https://pmc.ncbi.nlm.nih.gov/tools/developers/ Participants are responsible for complying with applicable article-level license and usage requirements. PMC identifies these services as the supported mechanisms for automated retrieval of PMC content.
- **PMC Cloud Service / AWS — Bulk Download**
    - The PMC Cloud Service provides a convenient mechanism for downloading the Challenge Corpus in bulk.
    - Available article files include JATS XML, plain text, metadata, and, where applicable, PDFs and supplementary materials.
    - The PMC datasets are accessible through AWS without requiring an AWS account or login.
    - Participants should use the provided PMCID list to identify and retrieve the articles comprising the PMC component of the Challenge Corpus.
    - For more information on the PMC Cloud Service and available datasets, please see: https://pmc.ncbi.nlm.nih.gov/tools/cloud/

- For general information about PubMed Central, please see: https://pmc.ncbi.nlm.nih.gov/

Submission Structure

Participants in this challenge will submit a submission file containing responses to up to 1,000 natural-language questions comprising the evaluation set. On January 15, 2027, participants must submit a Docker image(s) containing their solution, a description of their system, including its components, technical documentation, and deployment and execution instructions, in accordance with the System Submission requirements provided below.

Participants will be provided the evaluation set of questions on January 18, 2027. Subsequently, participants should develop responses to these questions and submit via the challenge portal prior to or on January 31, 2027. The participants can submit up to three (3) submissions for this task. Participants should follow this format and upload their submission files via the challenge portal.

Response submission files should be UTF-8-encoded and use JSONL file structure. Each JSONL record represents the response to one question. The metadata object identifies the participating team, system run, and question and contains the original request. The response object contains the system's generated answer, accompanied by supporting_evidence consisting of textual excerpts from the retrieved publication that support the system's answer.

Each JSONL record contains four top-level keys:

1. Metadata

Contains information identifying the submission and the question:

- **team_id** — identifier for the participating team.
- **submission_id** — identifier for the submission.
- **info_need_id** — unique identifier for the question.
- **info_need** — the natural-language question.

2. Response_str

Contains the complete response generated by the participating system.
The response should include:

- The answer to the question.
- Inline citation markers in square brackets, such as [1], [2], supporting each answer sentence.
- A references section identifying the sources used. Must contain valid PMCIDs /PMIDs and evidence excerpts.

3. Sentences

Contains a structured, sentence-level representation of the response.
Each sentence record contains:

- **text** — the text of the sentence, without the inline citation markers.
- **supported_citations** — an array containing the reference_id values of the references that support the sentence.

For example:

```
"sentences": [
 {
  "sentence1": {
   "text": "Ibrutinib inhibits Bruton's Tyrosine Kinase (BTK).",
   "supported_citations": ["1"]
  }
 },
 {
  "sentence2": {
   "text": "Mutations in BTK and PLCG2 have been associated with treatment resistance.",
   "supported_citations": ["2"]
  }
 }
]
```

This provides an explicit claim-to-reference mapping, which can be used to evaluate citation correctness and response grounding.

4. References

Contains the sources cited in the response.
Each reference contains:

- **reference_id** — the identifier used by the citation markers in the response and supported_citations.
- **citation** — the PMID or PMCID.
- **ref_excerpt** — an excerpt from the referenced source that supports the response. In the sample submission file provided below, this is represented by the placeholder: te*xt excerpt not included in this demo response.* In an actual

submission, this field would contain the relevant supporting passage from the source.

Overall structure

The resulting hierarchy is:

```
Submission Record
│
├── metadata
│   ├── team_id
│   ├── submission_id
│   ├── info_need_id
│   └── info_need
│
├── response_str
│
├── sentences
│   ├── sentence1
│   │   ├── text
│   │   └── supported_citations
│   ├── sentence2
│   │   ├── text
│   │   └── supported_citations
│   └── ...
│
└── references
    ├── reference_id
    ├── citation
    ├── ref_excerpt
    └── ...
```

Relationship between the Fields

The key relationships are: metadata.info_need → response_str → sentences → references

More specifically:

Question
↓
System Response
↓
Individual Claims/Sentences/Nuggets
↓
Supporting References
↓
Evidence Excerpts

For example, if sentence2 has: "supported_citations": ["2"]. then "2" corresponds to the reference in references where: "reference_id": "2"
The ref_excerpt for reference 2 provides the specific evidence supporting that claim. This structure therefore supports evaluation at three levels: overall response quality, sentence-level grounding/citation correctness, and reference-level evidence support at sentence and claim/nugget level.

A nugget is a distinct, atomic, independently assessable piece of information that directly contributes to answering a question. A nugget should express a single meaningful proposition rather than combining multiple facts, relationships, or conclusions. In the example given above, examples of nuggets include:

- Mutations in **BTK** were associated with clonal expansion in patients with relapsing CLL.
- The clonal expansion associated with PLCG2 mutations suggests acquisition of ibrutinib treatment resistance.

System Submission

By January 15, 2027, participants must submit a Docker image(s) containing the solution, a description of their system and its components, together with sufficient technical documentation and instructions to enable NIH to reproduce, deploy, and evaluate the system within an NIH-managed AWS environment. All software, models, indexes, data resources, documentation, and other materials that comprise or support the submitted system must have been created or last modified on or before January 15, 2027. A link to the participant's GitHub/Docker repository for the solution will be accepted.

Submissions should describe the system architecture and components, including retrieval, ranking, embeddings, indexes, models, LLM inference, and any other software or data resources required to execute the system. Participants may design their systems as a single containerized application or as multiple containerized services, including distributed architectures in which different components run on separate compute nodes. Participants should also provide deployment and execution instructions, along with an estimate of the computational and storage resources required (e.g., CPU, memory, GPU, and storage).

Participants will not be required to submit executable container images, source code, or large data artifacts (e.g., embeddings, indexes, or models) as part of this initial submission. These materials will be requested during the evaluation phase in accordance with the Challenge procedures and technical guidance.

For evaluation, NIH will provision and manage the underlying AWS infrastructure, including compute, storage, networking, and orchestration. Participants will not be required to provision or manage AWS resources, Kubernetes clusters, or Terraform infrastructure.

To support verification of the January 15, 2027 submission deadline, participants may be required to provide cryptographic hashes or other verifiable records for software, models, indexes, data artifacts, and other system components identified in their submission. When requested during the evaluation phase, the corresponding executable or data artifacts must match the versions documented and verified at the submission deadline.

Model and Resource Restrictions: Submitted systems may use only publicly available, open-weight Large Language Models (LLMs), open-source software, and/or other publicly available computing resources. Systems may not invoke external APIs to proprietary frontier models (e.g., commercial hosted LLM endpoints) or rely on non-public, restricted-access software, models, datasets, or other resources during execution. Any pretrained/fine-tuned models used by the system must be included within the submitted Docker images.

- Docker Image Execution and Network Restrictions
  - The challenge organizers reserve the right to run the submitted Docker-based system and verify system reproducibility, resource utilization, and compliance with challenge rules. Specifically:
  - External Calls: Submitted Docker Images will be executed in a completely isolated environment with all external network calls blocked. Limited access will be permitted to NLM-hosted services, such as the UMLS Terminology Services (UTS), and other challenge specific services. Details of these approved services and endpoints will be made available when the challenge is published. This zero-egress policy (excluding the permitted services) prevents both data leaks and runtime reliance on external computing resources, tools, or services.

Additional technical guidance, including submission formats, deployment specifications, resource limits, verification requirements, and evaluation procedures, will be provided to participants sufficiently in advance of the January 15, 2027 submission deadline.

Evaluation and Challenge Winner Selection

To determine the challenge winners for this task, participants will submit their developed systems and a submission file containing responses to approximately 1,000 natural-language task-specific questions comprising the evaluation set. Evaluation will be conducted in two phases.

Phase 1: System Selection

The evaluation dataset will be 200 questions selected for the initial system-selection phase. This phase will identify the top-performing systems for more extensive evaluation in Phase 2. The primary metric will be the Grounded Nugget F1, which measures the system's ability to provide responses that are both correct and complete with respect to reference response, and that are supported by evidence from the challenge corpus. Grounded Nugget F1 is defined as follows:

$$Grounded\ Nugget\ F1 = \frac{2 \times GNP \times GNR}{GNP + GNR}$$

Where:

- **Grounded Nugget Precision (GNP)** = proportion of nuggets in the system response that are correct (i.e. present in the reference response) and supported by provided evidence:

$$GNP = \frac{|Correct\ and\ evidence\ supported\ nuggets\ in\ system\ response|}{|Total\ nuggets\ in\ system\ response|}$$

  - **Plain-language description:** GNP equals the number of nuggets in the system response that are both correct and supported by evidence, divided by the total number of nuggets in the system response. *Note: This equation is built using Word's native equation format for accessibility reasons.*
- **Grounded Nugget Recall (GNR)** = proportion of reference nuggets that the system correctly covers with supporting evidence:

$$GNR = \frac{|reference_nuggets\ correctly\ covered\ with\ supporting\ evidence|}{|Total\ reference_standard\ nuggets|}$$

- ☐ **Plain-language description:** GNR equals the number of reference nuggets that are correctly covered with supporting evidence, divided by the total number of reference-standard nuggets. *Note: This equation is built using Word's native equation format for accessibility reasons.*

While Grounded Nugget F1 (GNF1) will be the primary response-quality metric, a secondary metric based on Conflict / Controversy accurate (CCA) detection will be folded in using a weighted sum. CCA will measure a system's ability to correctly identify and characterize conflicts, disagreements, or controversies associated with the question. The metric will be based on precision, recall, and F1, where:

- **CCA Precision** measures the proportion of conflicts or controversies identified by the system that are supported by the evaluation evidence. It is calculated as the number of correctly identified conflicts/controversies (true positives) divided by the total number of conflicts/controversies identified by the system (true positives + false positives).
- **CCA Recall** measures the proportion of conflicts or controversies present in the reference/evaluation set that are correctly identified by the system. It is calculated as the number of correctly identified conflicts/controversies (true positives) divided by the total number of conflicts/controversies that should have been identified (true positives + false negatives).
- **CCA F1** is the harmonic mean of CCA Precision and CCA Recall

The over Task 2 Quality Score (T2QS) is defined as follows:

$$Task2_Quality_Score = w \times \mathrm{GNF1} + (1 - \mathrm{w}) \times \mathrm{CCA}$$

Plain-language description: The Task 2 quality score equals w times GNF1, plus the quantity one minus w, times CCA. Here, w is a weight between 0 and 1 that controls the balance between the two component scores, GNF1 and CCA. *Note: This equation is built using Word's native equation format for accessibility reasons.*

A higher Task 2 Quality Score (T2QS) indicates better-performing systems. The top N systems based on Phase 1 performance will advance to Phase 2.

Phase 2: Extended Evaluation and System Deployment

In this phase, each of the semifinalist systems, in the form of a Docker container(s), will be installed and configured in an NIH-provided AWS cloud environment to ensure a standardized evaluation environment. These systems will be evaluated using the private set of questions, ease of system installation, resource usage (memory, disk, compute), and response time.

Phase 2 will use the following evaluation criteria:

- **Task 2 Quality Score (T2QS)** composed of:
  - **Grounded Nugget F1:** Measures the precision and recall of correct, evidence-grounded nuggets.
  - **Conflict / Controversy Coverage:** Measures whether systems appropriately identify and characterize conflicting findings, scientific uncertainty, competing hypotheses, or other areas of disagreement when present.
- **Ease of Use:** Assesses the effort required to deploy, configure, operate, and use the system in the standardized NIH cloud environment.
- **Compute Time:** Measures the computational time required to generate responses.
- **Compute Effort:** Measures the computational effort required to generate responses, including the resources and processing required by the system.
- **Memory Footprint:** Measures the memory and other computational resources required to operate the system and generate responses.

T2QS will serve as the primary metric for determining final system rankings and identifying the overall winning systems. In the event of a tie, the additional evaluation criteria listed above may be used to differentiate systems and determine the final ranking.